

Agentic Mobile QA — before you arrive

A 3-hour hands-on workshop. Everything runs locally on your machine: an Android emulator, Appium, and an LLM served by Ollama.

 Do this 4–5 days ahead

 No API key needed

 macOS · Linux · Windows + WSL2

 One command installs everything

Please finish this before the session — and verify it.

The setup downloads about 10 GB (an Android system image, a 4.9 GB language model, a JDK) and takes 20–40 minutes on good Wi-Fi. The 25-minute setup block on the day is for fixing stragglers, not first-time installs. If you arrive without the green report from §3, you will not be able to follow the hands-on labs.

1 Hardware & accounts

- ☐ **macOS** (Apple Silicon or Intel), **Linux**, or **Windows 11 + WSL2**
- ☐ **≥ 16 GB RAM** — the emulator and the 8B model both run locally. 8 GB will struggle.
- ☐ **~20 GB free disk** — Android system image, emulator, JDK, llama3.1 (4.9 GB), node_modules, the demo app
- ☐ A **GitHub account** — read access is enough (used in the CI chapter)

2 Install everything with one command

Open a terminal and paste this line. It installs — or repairs — the whole toolchain, clones the workshop repository to `~/droid-detective`, downloads the demo app and the model, and prints a report at the end.

```
curl -fsSL https://raw.githubusercontent.com/QA-Olympians-Academy/droid-detective/main/setup.sh | bash
```

macOS

Install

Homebrew

first if `brew --version` fails: paste the one-line command from

brew.sh

, then open a *new* terminal and run the command above. Nothing needs `sudo`.

Windows

Install WSL2: in an *administrator* PowerShell run `wsl --install`, reboot, open the

Ubuntu

app and run the command *there* — not in PowerShell. The emulator needs Windows 11 with nested virtualization (see §6).

Linux

Nothing to prepare. It asks for your `sudo` password once (Ollama's installer). The emulator needs `/dev/kvm` — on Ubuntu: `sudo apt install qemu-kvm` and add yourself to the `kvm` group.

What it installs (skipping anything you already have)

- Node.js 20+ and npm 10
- JDK 17
- Android SDK: platform-tools, emulator, platform 34, build-tools 34.0.0, system image
- The `workshop_avd` emulator (Pixel 6, API 34)
- Appium 3 with the UiAutomator2 driver
- AppClaw CLI (`@appclaw/cli`)
- Ollama, running as a background service, and the `llama3.1` model
- The repo, its dependencies, the demo APK and `.env`

Good to know

- **Re-run it any time.** Every step is idempotent — it only does what is still missing, so a failed download is fixed by running the same command again.
- **Already have Android Studio?** Its SDK at the default location is reused; only the missing packages are added.
- **Already have Node / a JDK?** Kept, as long as they are Node 20+ and JDK 17+.
- **Prefer to install by hand?** Follow §0 of `docs/PLAYBOOK-3H.md` in the repo, then finish with `./setup.sh --check`. Mind the two traps in §5.

- `JAVA_HOME` , `ANDROID_HOME` and `PATH` in your shell profile
- **Open a new terminal** when it finishes — the environment variables land in `~/.zshrc` or `~/.bashrc` .

3 The report — what green looks like

The installer ends with this report. **Every line must show ✓**. Screenshot it — that is your ticket into the hands-on labs. Re-print it any time with `cd ~/droid-detective && ./setup.sh --check` .

```

► Environment report
✓ Node.js                v24.20.0
✓ pnpm                  10.28.0
✓ JDK                   17.0.19   /opt/homebrew/opt/openjdk@17
✓ adb                   37.0.1-15733141
✓ emulator              version 37.1.11.0
✓ platform android-34   present
✓ build-tools 34.0.0    aapt2 present
✓ system image          android-34 google_apis arm64-v8a
✓ AVD workshop_avd      present
✓ Appium                 3.7.0
✓ UiAutomator2 driver   installed
✓ AppClaw CLI           2.7.0
✓ Ollama                 0.31.1 · server up
✓ model llama3.1        present
✓ repo                  /Users/you/droid-detective
✓ node_modules          installed
✓ apps/demo.apk         129M
✓ .env                  present

✓ All green. Screenshot this and bring it to the workshop.
Next: ./setup.sh --boot then pnpm test (from /Users/you/droid-detective)

```

A `x` line names what is missing and how to fix it; in almost every case the fix is simply to run the install command again. A `△` line is a warning you should read, not a blocker.

4 Boot the emulator and run the smoke test

This is the state you should arrive in: emulator up, the test suite installing the app and running.

```

cd ~/droid-detective
./setup.sh --boot      # starts workshop_avd and waits until Android has booted
pnpm test              # installs the demo app on the emulator and runs the specs

```

Green condition — `--boot` ends with `emulator-5554` device, and `pnpm test` **starts, installs the app, and runs specs**.

Some assertions *may fail* — that is fine and expected. The point is that the stack works end to end.

Leave the emulator running or close it — `./setup.sh --boot` is all you need on the day. On Linux inside WSL2 without `systemd` , start Ollama in a second terminal with `ollama serve` before running the report.

5 What gets installed, where — and the two traps

For reference, and for anyone installing by hand. The installer's report checks every row.

TOOL	VERSION	WHERE THE INSTALLER PUTS IT	VERIFY
Node.js · pnpm	20+ · 10	macOS: Homebrew <code>node@24</code> · Linux: <code>~/.local/node</code>	<code>node -v</code> · <code>pnpm -v</code>
JDK	17	macOS: <code>brew install openjdk@17</code> · Linux: <code>~/.local/jdk-17</code>	<code>java -version</code>
Android SDK	platform 34 · build-tools 34.0.0 · emulator · system image	macOS: <code>~/Library/Android/sdk</code> · Linux: <code>~/Android/Sdk</code> (an Android Studio SDK there is reused)	<code>adb --version</code> · <code>emulator --version</code>
Emulator AVD	Pixel 6 · API 34	<code>workshop_avd</code> — arm64-v8a on Apple Silicon, x86_64 elsewhere	<code>emulator --list-avds</code>
Appium + UiAutomator2	3.x	global npm package · driver in <code>~/.appium</code>	<code>appium driver list --installed</code>
AppClaw CLI	2.x	global npm package <code>@appclaw/cli</code>	<code>appclaw --version</code>
Ollama + llama3.1	—	macOS: Homebrew service · Linux: system service · model in <code>~/.ollama</code>	<code>ollama list</code>
Repo · deps · APK · .env	—	<code>~/droid-detective</code> (or the clone you run it from)	<code>./setup.sh --check</code>
Environment variables	—	a marked block in <code>~/.zshrc</code> / <code>~/.bashrc</code>	<code>echo \$ANDROID_HOME</code>

⚠ Two traps that cost people an evening if they install by hand

macOS JDK: do not use `brew install --cask temurin` — it runs a `.pkg` installer that needs `sudo`. Use `brew install openjdk@17` (a formula) and set `JAVA_HOME=$(brew --prefix openjdk@17)`.

AppClaw: install the scoped `@appclaw/cli` (2.x). The old unscoped `appclaw` package is deprecated and fails with `ETARGET / No matching version found for df-vision@1.1.79`.

6 🛠 If something breaks

SYMPTOM	FIX
Homebrew is required on macOS	Install it from <code>brew.sh</code> , open a new terminal, run the install command again.
<code>x</code> step failed: ...	Read the error printed just above it (usually a dropped download or a declined <code>sudo</code>), then run the same command again — it resumes where it stopped.
<code>bash: command not found / Unsupported OS</code> on Windows	You are in PowerShell or Git Bash. Open the Ubuntu (WSL2) app and run it there.
<code>no /dev/kvm</code> — emulator dead-slow or will not start	Linux: <code>sudo apt install qemu-kvm && sudo adduser \$USER kvm</code> , log out and in. WSL2: add <code>nestedVirtualization=true</code> under <code>[wsl2]</code> in <code>%UserProfile%\wslconfig</code> , then <code>wsl --shutdown</code> — or run the emulator in Android Studio on the Windows side.
<code>adb / emulator : command not found</code> in a new terminal	The variables were written to <code>~/.zshrc</code> or <code>~/.bashrc</code> , whichever your login shell uses. <code>source</code> it, or use <code>./setup.sh --boot</code> which does not need them.
Unknown AVD name <code>[workshop_avd]</code>	Run the install command again (it creates the AVD), or <code>./setup.sh --boot</code> .
<code>adb: no devices/emulators found</code>	Emulator not booted — <code>./setup.sh --boot</code> ; or <code>adb kill-server && adb start-server</code> .
Could not find 'aapt2'	build-tools 34.0.0 is missing — run the install command again.

SYMPTOM	FIX
Could not find a driver for 'UiAutomator2'	<code>export APPIUM_HOME=\$HOME/.appium</code> , then <code>appium driver install uiautomator2</code> .
Ollama hangs or is very slow	First load into RAM — run <code>ollama run llama3.1</code> once beforehand and close other apps.
AppClaw: <code>model 'llama3.2' not found</code>	You launched it outside the repo root, so <code>.env</code> was not loaded — run from <code>~/droid-detective</code> .



On the day, bring

Your laptop with the green report from §3 and the emulator booting · your charger · a GitHub login you can actually sign into. If you get stuck on any step, send the failing command and its full output ahead of the session — that is far quicker than debugging it in the room.